

WorldSkills ASEAN 2025 – Network Systems Administration

Day 1 – Network Foundation Rebuild

Date: June 3, 2026

Objective

Rebuild the lab according to the official WorldSkills topology using Debian 12.7. Deploy fw and int-srv01, configure IPv4 and IPv6 networking, and verify connectivity.

Topology

fw (Debian 12.7): Firewall, VPN, Proxy

int-srv01 (Debian 12.7): LDAP, CA, Samba, Internal DNS, DHCP with DDNS, Syslog

Firewall Configuration

ens18=WAN, ens19=INT, ens20=DMZ, ens21=CLT.

INT: 10.1.10.1/24

DMZ: 10.1.20.1/24

CLT: 10.1.40.1/24

Proxmox Bridges

vmbr0=WAN, vmbr10=INT, vmbr20=DMZ, vmbr40=CLT

int-srv01 Configuration

IPv4: 10.1.10.10/24, Gateway: 10.1.10.1, DNS: 10.1.10.10

IPv6: 2001:db8:1001:10::10/64

Verification

Successful ping from int-srv01 to fw (10.1.10.1). Routing table verified.

Lessons Learned

NetworkManager, nmcli, IPv4/IPv6 addressing, routing, Proxmox bridge design, and topology validation.

Issue Discovered

Original lab used OPNsense. Official topology requires a Debian 12.7 firewall, prompting a clean rebuild.

Next Session

LDAP, Certificate Authority (CA), user accounts, OU structure, and authentication testing.